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Bayesian Time-Series Econometrics

Monetary Policy Transmission in Poland: A Bayesian VAR Analysis

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1 Introduction

Understanding how monetary policy transmits to real economic activity and prices is one of the central questions in applied macroeconomics. For small open economies such as Poland, this question is particularly relevant given the active use of interest rate policy by the Narodowy Bank Polski (NBP) across several distinct monetary cycles—including the aggressive tightening of 2001, the near-zero rate environment of 2015–2021, and the rapid rate-hike cycle of 2021–2022 in response to post-pandemic inflation.

The 2021–2023 tightening cycle is particularly important for the present analysis. Beginning in October 2021, the NBP raised rates aggressively in response to inflation driven by energy price shocks and the economic consequences of the Russian invasion of Ukraine, lifting WIBOR 3M from near zero to a peak of approximately 7.3% by September 2022—a swing of over 700 basis points in under a year—while CPI inflation exceeded 15% in early 2023, its highest level in three decades. This abrupt and large-scale policy shift is one of the most informative episodes in the sample for studying the dynamic transmission of monetary conditions to output and prices.

The canonical framework for studying monetary transmission is the structural Vector Autoregression (VAR), introduced by Sims (1980). However, standard VAR models estimated by Ordinary Least Squares (OLS) face a well-known dimensionality problem: the number of freely estimated parameters grows quadratically with the number of variables and lags, leading to over-fitting in short macroeconomic samples. The Bayesian VAR (BVAR) framework, pioneered by Doan et al. (1984) and Litterman (1986), addresses this problem by introducing a *Minnesota Prior* that shrinks VAR coefficients towards a parsimonious random-walk representation, improving both in-sample fit and out-of-sample forecast performance.

This paper estimates a three-variable BVAR for Poland comprising GDP growth, CPI inflation, and the WIBOR 3-month interest rate. The model is estimated using a Gibbs sampler over 96 quarterly observations (2000Q1–2023Q4). Our primary object of interest is the Impulse Response Function (IRF) of CPI and GDP to an identified monetary policy shock, using the standard recursive (Cholesky) identification scheme.

The remainder of the paper is organised as follows. Section 2 describes the data and preliminary diagnostics. Section 3 outlines the econometric methodology. Section 4 presents the main results and robustness checks. Section 5 concludes.

2 Data

2.1 Variables and Sources

The analysis uses three quarterly time series for Poland over the period 2000Q1–2023Q4 ($T = 96$ observations), all sourced from Eurostat via its public REST API and summarised in Table 1.

Table 1: Variable definitions and data sources

Variable	Symbol	Definition	Source
GDP YoY (%)	<i>gdp_yoy</i>	Chain-linked real GDP, % change vs. same quarter of previous year	Eurostat: <i>namq_10_gdp</i>
CPI YoY (%)	<i>cpi_yoy</i>	HICP annual rate of change, monthly → quarterly average	Eurostat: <i>prc_hicp_manr</i>
WIBOR 3M (%)	<i>wibor_3m</i>	3-month Warsaw Interbank Offered Rate, monthly → quarterly average	Eurostat: <i>irt_st_m</i>

The choice to use WIBOR 3M rather than the NBP reference rate as the policy instrument deserves brief justification. WIBOR 3M is a market-determined rate that reflects actual interbank borrowing costs and responds to monetary policy expectations continuously: market participants price in anticipated future decisions, so WIBOR typically leads official rate changes at cyclical turning points. This forward-looking property makes it a richer and more timely signal of monetary conditions than the NBP reference rate, which moves only in discrete steps at scheduled Monetary Policy Council meetings. For time-series modelling at quarterly frequency, a continuous market price also fits more naturally into the dynamic assumptions of a VAR framework. We acknowledge that WIBOR 3M embeds a liquidity premium and a term risk premium on top of the pure policy rate component, so it is not a clean instrument for measuring policy intentions alone. Nevertheless, for capturing aggregate monetary conditions as they actually feed through to borrowing costs across the economy, WIBOR 3M is the more economically relevant measure.

The sample spans four distinct macroeconomic episodes: the high-inflation, high-interest-rate environment of the early 2000s; the global financial crisis of 2008–2009; the COVID-19 recession of 2020; and the energy-price-driven inflation surge of 2021–2023. This variation across monetary cycles is important for the identification of monetary policy shocks. Figure 1 plots all three series.

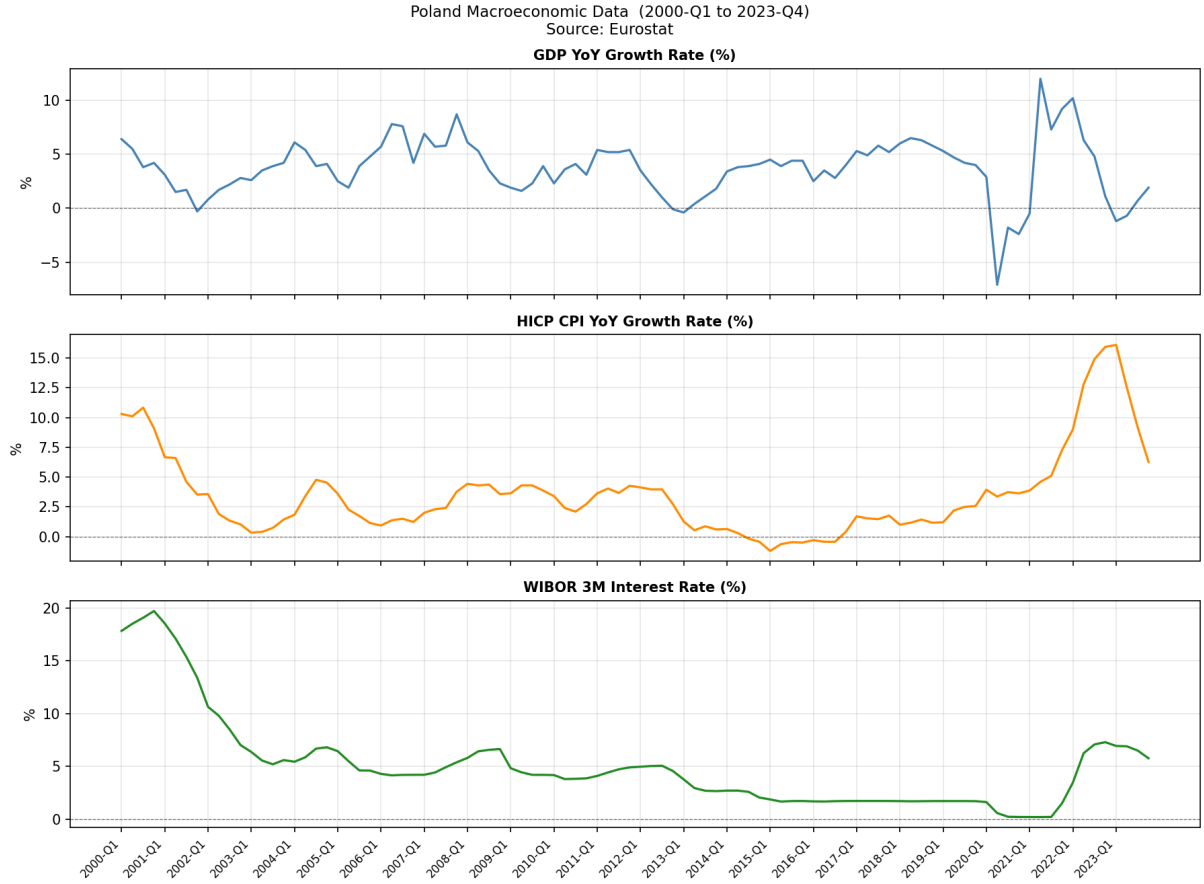


Figure 1: Poland macroeconomic data, 2000Q1–2023Q4. *Top:* GDP YoY growth rate (%). *Middle:* HICP CPI YoY growth rate (%). *Bottom:* WIBOR 3-month interest rate (%). Source: Eurostat.

2.2 Preliminary Diagnostics

Augmented Dickey–Fuller (ADF) unit-root tests indicate mixed orders of integration: GDP growth and WIBOR 3M are stationary at the 5% level in levels ($p = 0.0002$ and $p = 0.0012$, respectively), while CPI inflation requires first differencing to achieve stationarity.

Given the mixed evidence, we additionally conducted a Johansen (1991) cointegration test with a restricted constant (`det_order = 0`) and two autoregressive lags. The trace statistic clearly rejects the null of no cointegration (trace statistic: 53.52, 95% critical value: 29.80). However, the full-rank result ($r = 3$) is more naturally interpreted as evidence that the system as a whole behaves in a stationary manner that is consistent with the ADF finding that two of the three variables are already $I(0)$ —rather than as evidence of a conventional cointegrating relationship among three $I(1)$ variables. These results support estimation in levels. Following Sims (1980), who argues that level-form VARs yield consistent impulse-response estimates regardless of the integration order of individual variables, we estimate the BVAR in levels with an intercept, preserving dynamic

information without imposing potentially incorrect differencing restrictions.

3 Methodology

3.1 The VAR Model

Let $\mathbf{Y}_t = (\text{GDP}_t, \text{CPI}_t, \text{WIBOR}_t)^\top$ denote the (3×1) vector of endogenous variables at time t . The reduced-form VAR(p) is:

$$\mathbf{Y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{Y}_{t-1} + \mathbf{A}_2 \mathbf{Y}_{t-2} + \cdots + \mathbf{A}_p \mathbf{Y}_{t-p} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}), \quad (1)$$

where $\mathbf{c}$ is a (3×1) intercept vector, $\mathbf{A}_j$ are (3×3) coefficient matrices for $j = 1, \dots, p$, and $\mathbf{\Sigma}$ is the (3×3) residual covariance matrix. Based on the Bayesian Information Criterion (BIC), we set $p = 2$, yielding $1 + 3 \times 2 = 7$ free parameters per equation and $3 \times 7 = 21$ parameters across the full system, giving a per-equation parameter-to-observation ratio of $96/7 \approx 13.7$ and a system-wide ratio of $96/21 \approx 4.6$.

3.2 Minnesota Prior

The Minnesota Prior (Litterman, 1986; Doan et al., 1984) encodes three economic beliefs about the data-generating process:

1. **Random-walk prior.** Each variable is likely well approximated by a random walk, so the prior mean for own first-lag coefficients is one, and all other lag coefficients are centred at zero.
2. **Lag decay.** More distant lags carry less information, so prior variances shrink at rate $1/l^2$ for lag l .
3. **Cross-variable shrinkage.** Coefficients on other variables' lags are less informative than own-lag coefficients, so cross-variable prior variances are additionally scaled by $\kappa^2 < 1$.

A natural question is whether the random-walk prior mean of one is appropriate for variables expressed as growth rates (GDP YoY) and inflation rates (CPI YoY) rather than log-levels. While the random-walk assumption is most literally correct for log-levels, Litterman (1986) and subsequent applied work (e.g. Banbura et al. 2010) have shown that the Minnesota Prior remains an effective regulariser for stationary-ish growth-rate variables even when the random-walk assumption is not strictly satisfied: the key role of the prior is shrinkage rather than literal belief in unit roots. Consistent with this view, our posterior estimates for GDP and CPI own-lag coefficients are well below one (0.695 and net 0.954, respectively), confirming that the data pulls the posterior away from the random-walk prior toward more mean-reverting dynamics.

Formally, for coefficient $\beta_{ij,l}$ (effect of variable j at lag l on variable i):

$$\text{Var}(\beta_{ij,l}) = \begin{cases} \left(\frac{\lambda}{l}\right)^2 & \text{if } i = j, \\ \left(\frac{\lambda \kappa}{l}\right)^2 \frac{\hat{\sigma}_i^2}{\hat{\sigma}_j^2} & \text{if } i \neq j, \end{cases} \quad (2)$$

where λ is the overall tightness parameter, $\hat{\sigma}_i^2$ are equation-specific residual variances estimated by OLS, and we set $\kappa = 0.5$. The baseline uses $\lambda = 0.2$; robustness checks consider $\lambda \in \{0.1, 0.4\}$.

3.3 Posterior Inference via Gibbs Sampler

Under the Minnesota Prior and a Normal–Inverse–Wishart conjugate structure for Σ , the conditional posteriors of the VAR coefficients and the covariance matrix are available in closed form, enabling efficient Gibbs sampling (Gelman et al., 2013).

Each Gibbs iteration proceeds in two steps. In the first step, the full vector of stacked VAR coefficients β is drawn from its conditional posterior, which is multivariate Normal: given the current draw of Σ and the data, the posterior mean and covariance of β are computed analytically by combining the Minnesota Prior with the OLS likelihood, and a single draw is taken from $\mathcal{N}(\bar{\beta}, \bar{V}_\beta)$. In the second step, the residual covariance matrix Σ is drawn from its conditional posterior, which is Inverse–Wishart: given the current draw of β , the VAR residuals are computed, the posterior scale matrix and degrees of freedom are updated, and a draw is taken from $\mathcal{IW}(\bar{S}, \bar{\nu})$. These two steps alternate for each iteration of the sampler.

We run two independent chains from overdispersed starting values, discarding the first 1 000 draws as burn-in and retaining 5 000 post-burn-in draws per chain (10 000 in total). The burn-in period allows the chain to travel from its arbitrary initial values to the high-probability region of the posterior, after burn-in, the retained draws are treated as dependent Monte Carlo draws from the joint posterior and are used to approximate posterior summaries. These retained draws are used directly to compute posterior means, credible intervals, and impulse response functions by evaluating each quantity of interest at every stored draw and summarising the resulting empirical distribution. Convergence is assessed using the Gelman–Rubin (1992) potential scale reduction factor $\hat{R}$.

3.4 Identification of Monetary Policy Shocks

We identify monetary policy shocks using the recursive (Cholesky) ordering $\mathbf{Y}_t = (\text{GDP}_t, \text{CPI}_t, \text{WIBOR}_t)^\top$. This ordering implies that the interest rate responds contemporaneously to output and inflation, but output and inflation do not respond to interest-rate shocks within the same quarter—the standard assumption in the monetary VAR literature

(Christiano et al., 1999). IRFs and their posterior credible bands are computed over a 20-quarter horizon.

4 Results

4.1 MCMC Convergence Diagnostics

The Gibbs sampler converges rapidly and reliably. The Gelman–Rubin $\hat{R}$ statistic is below 1.001 for all $3 \times 7 = 21$ VAR coefficient parameters (minimum: 0.9999; maximum: 1.0003; mean: 1.0000), well within the conventional threshold of 1.1. Trace plots confirm that the two chains mix thoroughly throughout the post-burn-in period, with no evidence of slow mixing or non-convergence.

4.2 Posterior Coefficient Estimates

Table 2 reports the posterior mean, standard deviation, and 68% credible interval for selected coefficients.

Table 2: Selected posterior coefficient estimates, BVAR(2), $\lambda = 0.2$. The 16% and 84% quantiles correspond to the one-standard-deviation posterior credible interval.

Regressor	Post. Mean	Post. Std	16%	84%	Note
<i>Panel A: Equation GDP YoY</i>					
GDP_{t-1}	0.695	0.088	0.608	0.782	Strong persistence
CPI_{t-1}	−0.256	0.170	−0.425	−0.086	Negative, credible
$WIBOR_{t-1}$	−0.038	0.301	−0.339	0.266	Wide CI, uncertain
<i>Panel B: Equation CPI YoY</i>					
CPI_{t-1}	1.149	0.069	1.081	1.218	High persistence
GDP_{t-1}	0.076	0.036	0.040	0.112	Demand pressure
$WIBOR_{t-1}$	0.057	0.123	−0.065	0.181	Uncertain
<i>Panel C: Equation WIBOR 3M (Policy Rule)</i>					
GDP_{t-1}	0.076	0.020	0.057	0.095	Tightening on growth
CPI_{t-1}	0.111	0.038	0.074	0.149	Tightening on inflation
$WIBOR_{t-1}$	1.122	0.069	1.054	1.190	Rate persistence

Several features stand out. In the GDP equation, lagged GDP growth carries a posterior mean of 0.695 with a credible interval entirely above zero, reflecting strong output persistence. CPI inflation at lag 1 enters with a negative coefficient (−0.256), consistent with stagflation dynamics reducing real activity. In the CPI equation, own-lag persistence is very high (1.149 at lag 1, partially offset by −0.195 at lag 2), consistent with the inertial nature of Polish inflation. Critically, in the WIBOR equation, both GDP growth (0.076) and CPI inflation (0.111) carry positive and credible coefficients, suggesting

that the policy-rate equation is consistent with a Taylor-rule-type response to output growth and inflation.

4.3 Impulse Response Functions

Effect on CPI Inflation

Figure 2 displays the IRF of CPI inflation to a one-standard-deviation contractionary WIBOR shock. The Cholesky decomposition implies a shock of approximately 0.45 percentage points to the WIBOR 3M rate, which corresponds to a moderate but realistic monetary tightening within a single quarter. The response exhibits the characteristic hump-shaped pattern documented in the empirical monetary literature. In the first two quarters following the shock, the response is slightly positive—a “price puzzle” commonly found in three-variable recursive VAR models (Sims, 1992). This artefact likely reflects omitted variables such as commodity prices or the exchange rate, which the NBP may respond to even as domestic inflation remains temporarily elevated; it does not invalidate the medium-run results. From the third quarter onward, the response turns negative.

The key finding is the posterior credibility of the disinflationary effect. Reporting 68% credible bands is standard practice in the Bayesian VAR literature (Sims and Zha, 1998) and corresponds to the one-posterior-standard-deviation interval. From horizon 5 (five quarters after the shock), the 68% posterior credible band lies entirely below zero. The trough response of -0.196 percentage points is reached at horizon 11, approximately three years after the initial shock. The 90% credible band intersects zero at around horizon 7, indicating that the effect is robust at the 68% level but associated with non-negligible uncertainty at wider probability coverage.

The slow and persistent nature of monetary transmission, which peaks after nearly three years—is consistent with the empirical literature on small open economies and reflects the gradual pass-through of interest rate changes to domestic demand, credit conditions, and ultimately wages and prices.

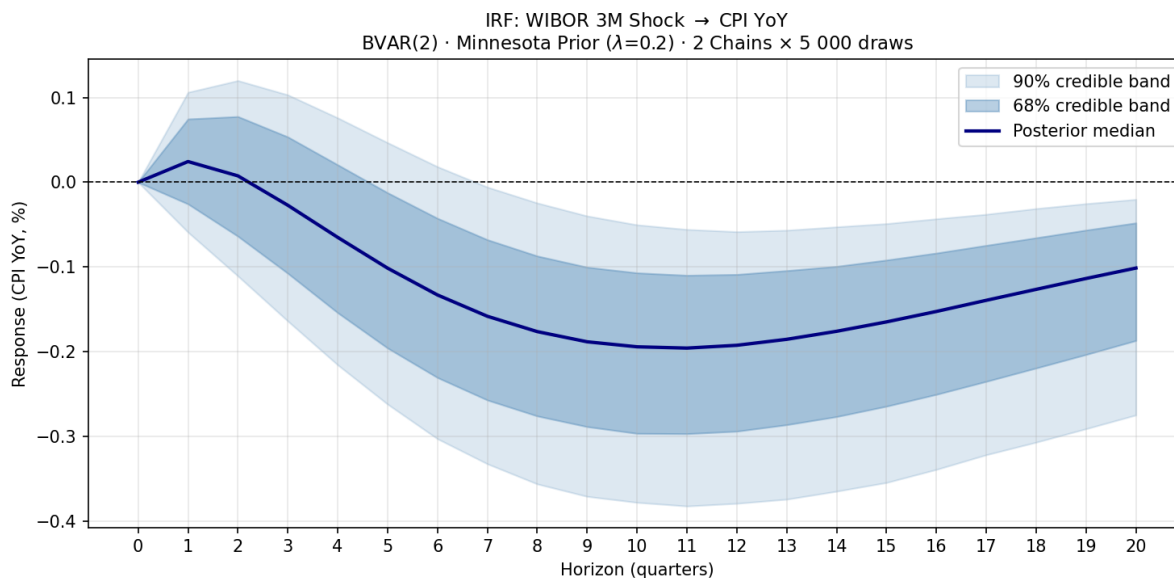


Figure 2: Impulse response function: WIBOR 3M shock $\rightarrow$ CPI YoY. BVAR(2), Minnesota Prior ($\lambda = 0.2$), Gibbs sampler ($2 \times 5,000$ draws). Dark band: 68% posterior credible interval. Light band: 90% posterior credible interval. Solid line: posterior median.

Effect on GDP Growth

Figure 3 shows the IRF of GDP growth to the same WIBOR shock. The posterior median response is negative throughout the 20-quarter horizon, with a trough of approximately -0.064 percentage points at horizon 4. The response is directionally consistent with theory and higher interest rates slow economic activity, but it is associated with substantially wider posterior uncertainty than the CPI response. The 68% credible band crosses zero between horizons 2 and 5, and the 90% band includes zero throughout.

The wider uncertainty around the GDP response relative to the CPI response is a common finding in the Bayesian VAR literature and reflects the greater noise in quarterly output data relative to price indices. It does not contradict the hypothesis of monetary transmission; rather, it indicates that the output channel, while present on average, is estimated with less precision given the available sample.

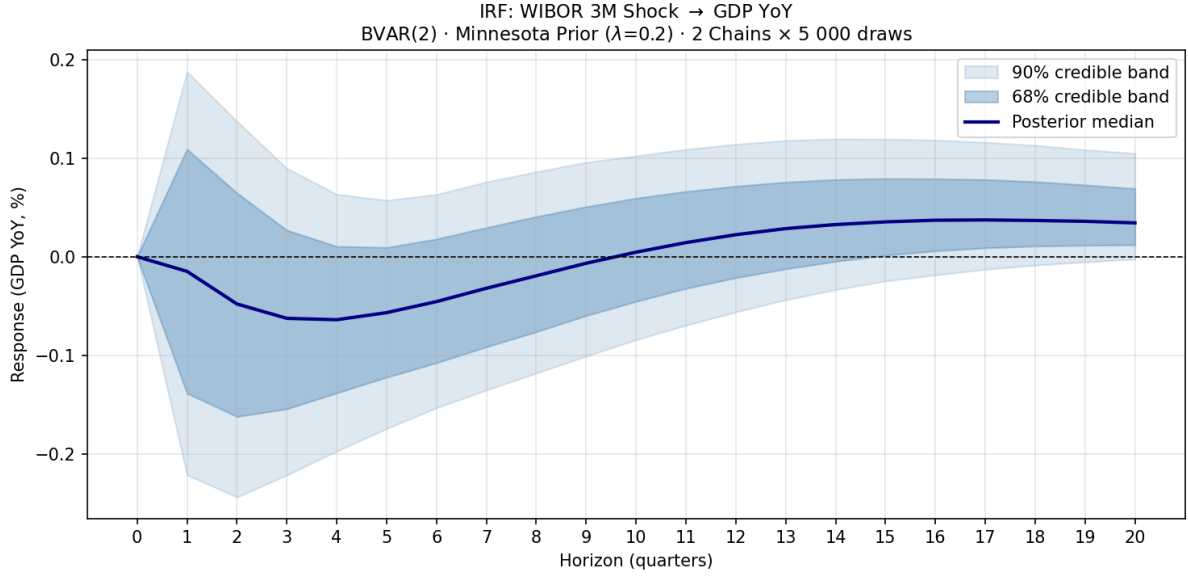


Figure 3: Impulse response function: WIBOR 3M shock → GDP YoY. BVAR(2), Minnesota Prior ($\lambda = 0.2$), Gibbs sampler ($2 \times 5,000$ draws). Shaded bands and line as in Figure 2.

4.4 Robustness: Prior Sensitivity

Figure 4 presents robustness checks for the CPI IRF across three values of the Minnesota Prior tightness parameter: $\lambda = 0.1$ (tight), $\lambda = 0.2$ (baseline), and $\lambda = 0.4$ (loose). Table 3 summarises the key statistics.

Table 3: Robustness of the CPI IRF to Minnesota Prior tightness.

λ	Peak horizon	IRF trough	68% CI at $h = 11$	68% band < 0 from
0.1 (tight)	$h = 11$	-0.202	$[-0.294, -0.126]$	$h = 3$
0.2 (baseline)	$h = 11$	-0.197	$[-0.298, -0.108]$	$h = 5$
0.4 (loose)	$h = 12$	-0.184	$[-0.295, -0.088]$	$h = 8$

The three IRF medians are remarkably similar: the peak responses are -0.202, -0.197, and -0.184, a maximum difference of $0.202 - 0.184 = 0.018$ percentage points. The credible bands substantially overlap across all three specifications. This prior insensitivity is an important finding: it implies that the disinflationary effect of monetary tightening is driven by the data rather than by prior regularisation, lending additional credibility to the baseline results.

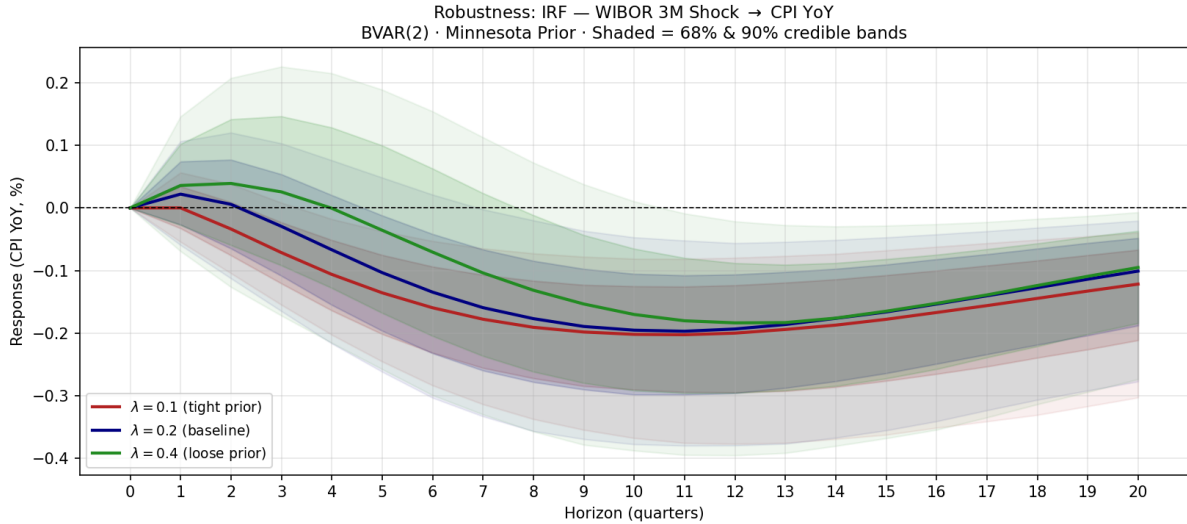


Figure 4: Prior sensitivity: IRF of CPI YoY to WIBOR 3M shock under three Minnesota Prior tightness values ($\lambda \in \{0.1, 0.2, 0.4\}$). Shaded regions show 68% and 90% posterior credible bands for the baseline ($\lambda = 0.2$). The three posterior medians are nearly identical, confirming robustness to prior choice.

5 Conclusion

This paper has estimated a Bayesian VAR model of monetary policy transmission in Poland using quarterly data from 2000 to 2023. The Minnesota Prior, implemented via a Gibbs sampler with two chains and full MCMC convergence diagnostics (all $\hat{R} < 1.001$), provides effective regularisation given the moderate sample size.

The central finding is that contractionary monetary policy, identified as an unexpected increase in the WIBOR 3-month rate, produces a statistically credible and moderate disinflationary effect on CPI inflation, with the 68% posterior credible band lying below zero from the fifth quarter onwards and reaching a trough of approximately -0.20 percentage points around the eleventh quarter. The GDP response is directionally consistent with theory but is estimated with greater uncertainty, a common feature of empirical monetary VAR models.

Robustness checks confirm that the disinflationary effect is insensitive to the choice of prior tightness across a wide range of λ values (0.1 to 0.4), indicating that the finding is data-driven rather than prior-induced.

These results have two implications. First, they are consistent with the view that NBP monetary tightening contributed to disinflation over the 2000–2023 period, though the three-variable Cholesky-identified BVAR cannot provide definitive causal evidence of policy effectiveness. Second, they illustrate the practical value of the Bayesian approach: the posterior credible bands deliver a complete and honest description of parameter uncertainty that frequentist confidence intervals—derived under asymptotic approximations—cannot

fully capture in samples of this size.

Future work could extend the analysis by incorporating additional variables (e.g., the exchange rate or credit growth), using sign restrictions for identification, or applying time-varying parameter models to capture potential structural changes in the transmission mechanism across monetary regimes. The present robustness analysis focuses on prior tightness; future extensions should additionally examine lag-length sensitivity and alternative Cholesky orderings.

References

- Banbura, M., Giannone, D., and Reichlin, L. (2010). Large Bayesian vector auto regressions. *Journal of Applied Econometrics*, 25(1), 71–92.
- Christiano, L. J., Eichenbaum, M., and Evans, C. L. (1999). Monetary policy shocks: What have we learned and to what end? In J. B. Taylor and M. Woodford (Eds.), *Handbook of Macroeconomics*, Vol. 1A, pp. 65–148. Elsevier.
- Doan, T., Litterman, R., and Sims, C. A. (1984). Forecasting and conditional projection using realistic prior distributions. *Econometric Reviews*, 3(1), 1–100.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis* (3rd ed.). CRC Press.
- Gelman, A. and Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences. *Statistical Science*, 7(4), 457–472.
- Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580.
- Litterman, R. B. (1986). Forecasting with Bayesian vector autoregressions: Five years of experience. *Journal of Business and Economic Statistics*, 4(1), 25–38.
- Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1–48.
- Sims, C. A. (1992). Interpreting the macroeconomic time series facts: The effects of monetary policy. *European Economic Review*, 36(5), 975–1000.
- Sims, C. A. and Zha, T. (1998). Bayesian methods for dynamic multivariate models. *International Economic Review*, 39(4), 949–968.
- Zellner, A. (1971). *An Introduction to Bayesian Inference in Econometrics*. Wiley.

A Python Implementation Notes

The model was implemented from scratch in Python, with `numpy` and `scipy` as the only core numerical dependencies. The Gibbs sampler was written without any dedicated Bayesian or VAR package: each iteration manually computes the posterior Normal parameters for the VAR coefficients and the posterior Inverse-Wishart parameters for the covariance matrix, then calls `numpy.random.multivariate_normal` and `scipy.stats.invwishart.rvs` respectively. The two chains were initialised from overdispersed starting values—the first chain used the OLS residual variance matrix as the initial $\hat{\Sigma}$, the second used twice that value—to facilitate convergence diagnosis via $\hat{R}$. All MCMC draws were stored as three-dimensional `numpy` arrays (draws $\times$ equations $\times$ variables), and impulse response functions were computed by recursively raising the VAR companion matrix to successive powers and extracting the relevant row-column element at each horizon. The full replication code is available at <https://github.com/wmf2241/bvar-poland>.